

4.2 Forces: Density & Pressure

Question Paper

Course	CIE A Level Physics
Section	4. Forces, Density & Pressure
Topic	4.2 Forces: Density & Pressure
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The formula for hydrostatic pressure is $P = \rho gh$

Which equation, or principle of physics, is used in the derivation of this formula?

- A. density = mass $\div$ volume
- B. potential energy = mgh
- C. atmospheric pressure decreases with height
- D. density increases with depth

[1 mark]

Question 2

The density of mercury is $13.6 \times 10^3 \text{ kg m}^{-3}$

The pressure difference between the bottom and the top of a column of mercury is 100 kPa.

What is the height of the column?

- A. 0.75 m
- B. 1.3 m
- C. 7.4 m
- D. 72 m

[1 mark]

Question 3

An object, immersed in a liquid in a tank, experiences an upthrust.

What is the physical reason for this upthrust?

- A. the density of the body differs from that of the liquid
- B. the density of the liquid increases with depth
- C. the pressure in the liquid increases with depth
- D. the value of g in the liquid increases with depth

[1 mark]

Question 4

The Mariana Trench in the Pacific Ocean has a depth of about 10 km.

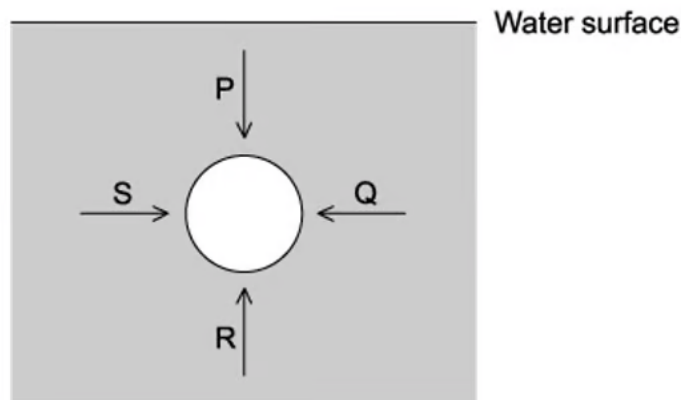
Assuming that sea water is incompressible and has a density of about 1020 kg m^{-3} , what would be the approximate pressure at that depth?

- A. 10^5 Pa
- B. 10^6 Pa
- C. 10^7 Pa
- D. 10^8 Pa

[1 mark]

Question 5

The diagram represents a sphere under water. P , Q , R and S are forces acting on the sphere, due to the pressure of the water.



Each force acts perpendicularly to the sphere's surface. P and R act in opposite directions vertically. Q and S act in opposite directions horizontally.

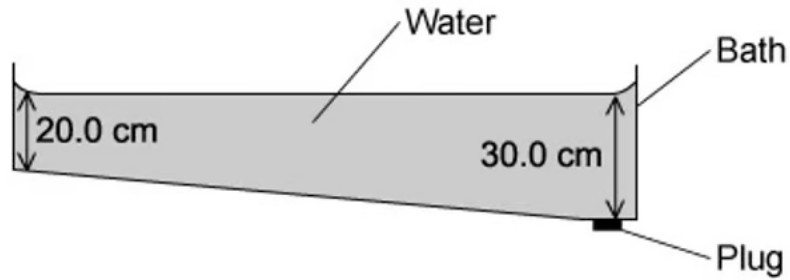
Which information about the magnitudes of the forces is correct?

- A. $P < R$ and $S = Q$
- B. $P > R$ and $S = Q$
- C. $P = R$ and $S = Q$ and $P \neq S$
- D. $P = R$ and $S = Q$ and $P = S$

[1 mark]

Question 6

Water in a bath varies in depth from 20.0 cm at the shallow end to 30.0 cm at the end with the plug.



The density of the water is 1000 kg m^{-3}

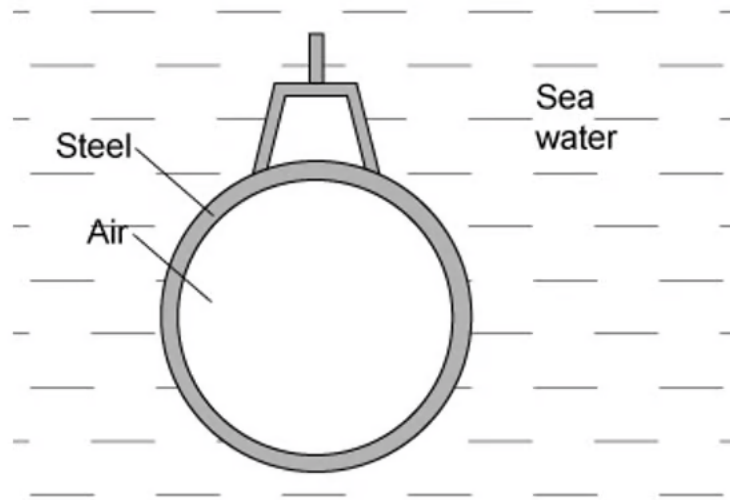
What is the pressure of the water acting on the plug?

- A. 0.98 kPa
- B. 2.0 kPa
- C. 2.9 kPa
- D. 290 kPa

[1 mark]

Question 7

A submarine is in equilibrium in a fully submerged position.



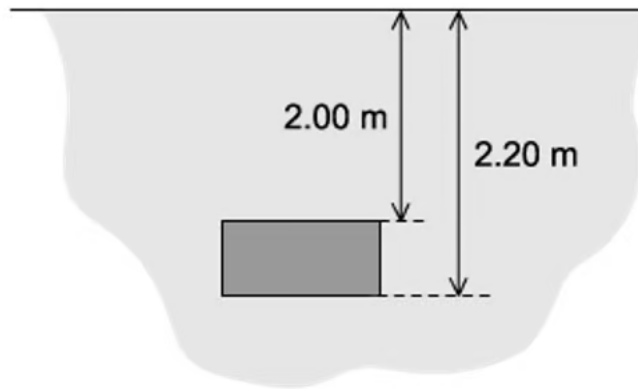
What causes the upthrust on the submarine?

- A. the air in the submarine is less dense than sea water
- B. the sea water exerts a greater upward force on the submarine than the weight of the steel
- C. the submarine displaces its own volume of sea water
- D. there is a difference in water pressure acting on the top and on the bottom of the submarine

[1 mark]

Question 8

The diagram shows a rectangular block of mass 8.2 kg immersed in sea water of density $1.10 \times 10^3 \text{ kg m}^{-3}$



What is the difference in pressure between the top and bottom surfaces of the block?

- A. $2.2 \times 10^2 \text{ Pa}$
- B. $2.2 \times 10^3 \text{ Pa}$
- C. $1.8 \times 10^4 \text{ Pa}$
- D. $2.3 \times 10^4 \text{ Pa}$

[1 mark]

Question 9

A horizontal plate of area 0.036 m^2 is beneath the surface of a liquid of density 930 kg m^{-3} . The force on the plate due to the pressure of the liquid is 290 N .

What is the depth of the plate beneath the surface of the liquid?

- A. 0.88 m
- B. 1.13 m
- C. 8.7 m
- D. 9.1 m

[1 mark]

Question 10

A rectangular metal bar exerts a pressure of 15 200 Pa on the horizontal surface on which it rests.

If the height of the metal bar is 80 cm, what is the density of the metal?

- A. 19 kg m^{-3}
- B. 190 kg m^{-3}
- C. 1900 kg m^{-3}
- D. $19\,000 \text{ kg m}^{-3}$

[1 mark]



Model Answers: Easy

1

The correct answer is **A** because:

- If you run through the beginning steps of the derivation, you will remember the two equations you start with are
 - $W = mg$
 - $\rho = \frac{m}{V}$
- In an MCQ you would not be required to carry out a derivation, but you will need to know it for the written paper. The full derivation is as follows:

Consider a column of water which occupies a total volume V , has a height h , and a base surface area of A . The weight of all the water is :

$$W = mg$$

The water has a density which is equal to $\rho = \frac{m}{V}$ and therefore $m = \rho V$, hence combining these two equations gives:

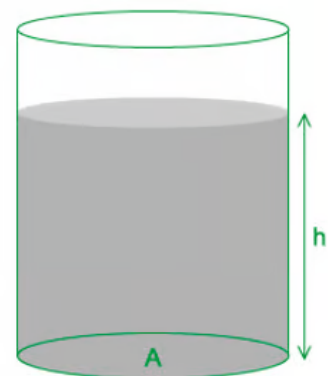
$$W = \rho Vg$$

For a cylinder, the volume is a product of its base surface area and height:

$$W = \rho Ahg$$

The water has a pressure which is equal to $P = \frac{F}{A}$ and therefore $F = W = PA$, hence combining these two equations gives:

$$P = \frac{\rho Ahg}{A} = \rho gh$$



2

The correct answer is **A** because:

- The pressure difference can be established from the equation of hydrostatic pressure
 - $\Delta P = \rho g(\Delta h)$
- Density of mercury, $\rho = 13\,600\text{ kg m}^{-3}$
- Acceleration due to gravity, $g = 9.81\text{ m s}^{-2}$
- Change in pressure, $\Delta P = 100\,000\text{ Pa}$
 - $\Delta h = \frac{\Delta P}{\rho g} = \frac{100\,000}{13\,600 \times 9.81} = 0.75\text{ m}$
- The height of the column is 0.75 m

3

The correct answer is **C** because:

- The physical reason for this upthrust is due to the difference in pressures at the top and the bottom of the object
- If we look at the equation
 - $P = \rho gh$
- We can deduce if the depth h increases, the pressure P increases too
- Therefore, the pressure at the bottom of the object is greater than the pressure at the top of the object
- The pressure acts on the surface area on the bottom of the object is the upthrust
- Upthrust is a force and this is directly related to pressure by the equation
 - $P = \frac{F}{A}$
- Therefore, the larger the pressure, the larger the force (upthrust) that is exerted on the object

A is incorrect because

while this could be true, this is not the physical reason for

B & D are incorrect because

the density ρ of a liquid remains constant, as does the volume of the surface

4

The correct answer is **D** because:

- The pressure difference can be established from the equation of hydrostatic pressure
 - $\Delta P = \rho g(\Delta h)$
- Density of sea water, $\rho = 1020 \text{ kg m}^{-3}$
- Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- Depth, $\Delta h = 10\,000 \text{ m}$
 - $\Delta P = \rho g(\Delta h) = 1020 \times 9.81 \times 10\,000 = 1.0 \times 10^8 \text{ Pa}$
- The approximate pressure at a depth of 10 km is $\sim 10^8 \text{ Pa}$

5

The correct answer is **A** because:

- Let's consider the horizontal forces Q and S initially
- These forces act at the same depth
- Pressure stays the same at the same depth so the magnitude of Q and S will be the same i.e.
 - $S = Q$
- To determine the relationship between forces P and R we must look at the equation
 - $P = \rho gh$
- We can deduce as the depth h increases, the pressure P increases too
- Therefore, the pressure at the bottom of the sphere is greater than the pressure at the top of the sphere
- Pressure is directly proportional to force so the force on the bottom of the sphere (R) will be greater than the force at the top of the sphere (P) i.e.
 - $P < R$

B is incorrect because

this implies the pressure is greater nearer the surface (C) which is not correct

C & D are incorrect because

these imply there is no difference in pressure at different depths in water which is also not correct

6

The correct answer is **C** because:

- The pressure on the plug can be established from the equation of hydrostatic pressure
 - $\Delta P = \rho g(\Delta h)$
- Density of water, $\rho = 1000 \text{ kg m}^{-3}$
- Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- Depth of plug, $\Delta h = 30 \text{ cm} = 0.3 \text{ m}$
 - $\Delta P = \rho g(\Delta h) = 1000 \times 9.81 \times 0.3 = 2943 \text{ Pa}$
- The pressure acting on the plug = 2.9 kPa (2 s. f.)

7

The correct answer is **D** because:

- Upthrust occurs because the pressure in a fluid increases with increasing depth
- The pressure acting at the bottom of the submarine is greater than the pressure acting at the top
- Pressure is directly proportional to force so the force on the bottom of the submarine will be greater than the force at the top of the submarine
- The resultant pressure causes a resultant upward force on the submarine, this is the upthrust

- A** is incorrect because the air in the submarine is less dense than sea water, however, upthrust is caused by pressure not by density
- B** is incorrect because the submarine is in equilibrium so the upthrust should be equal to the weight
- C** is incorrect because the volume of water displaced by the submarine does not influence the upthrust

8

The correct answer is **B** because:

- The pressure difference can be established from the equation of hydrostatic pressure

- $\Delta P = \rho g(\Delta h)$
- Density of sea water, $\rho = 1100 \text{ kg m}^{-3}$
- Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- Difference in depth, $\Delta h = 2.2 - 2.0 = 0.2 \text{ m}$
 - $\Delta P = \rho g(\Delta h) = 1100 \times 9.81 \times 0.2 = 2158 \text{ Pa}$
- The pressure difference between the top and bottom = $2.2 \times 10^3 \text{ Pa}$ (2 s. f.)

9

The correct answer is **A** because:

- The depth of an object in a liquid can be established from the equation of hydrostatic pressure
 - $P = \rho gh$
- Pressure is related to force by the equation $P = \frac{F}{A}$
 - $P = \rho gh = \frac{F}{A}$
 - $h = \frac{F}{\rho g A}$
- Density of liquid, $\rho = 930 \text{ kg m}^{-3}$
- Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- Force on plate, $F = 290 \text{ N}$
- Area of plate, $A = 0.036 \text{ m}^2$
 - $h = \frac{F}{\rho g A} = \frac{290}{930 \times 9.81 \times 0.036} = 0.88 \text{ m}$
- The depth of the plate beneath the surface of the liquid is 0.88 m

10

The correct answer is **C** because:

- The density of an object in a liquid can be determined from the equation of hydrostatic pressure
 - $P = \rho gh$
- Pressure, $P = 15\,200 \text{ Pa}$
- Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- Height of object, $h = 80 \text{ cm} = 0.8 \text{ m}$

$$\circ \quad \rho = \frac{P}{gh} = \frac{15\,200}{9.81 \times 0.8} = 1937 \text{ kg m}^{-3}$$

- The density of the metal is 1900 kg m^{-3} (2 s. f.)